

Peyton Hall

Assignment 3: Data Model to Database Design (Forward Engineering) Using Oracle Data Modeler

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The data model represents a simplified portion of the periodic table. There are three entities, i.e. Element, Category, and Period. Firstly, Element stores the following information about chemical elements: including `atomic_number`, `element_name`, `symbol`, `atomic_mass`, `group_number`, and references to category and period. Secondly, Category represents classifications of elements, which could be metal, nonmetal, or noble gas, and contains a unique `category_id`. Lastly, Period represents the horizontal rows of the periodic table and contains a unique `period_id`. Element records can be organized by both chemical classification and periodic table position according to these entities.

Two relationships are defined in the model. The first relationship connects Category to Element as one-to-many because each element belongs to exactly one category, while a category can contain many elements. The second relationship connects Period to Element as one-to-many because each element belongs to one period, while a period may contain many elements. Since these relationships represent the structure of the periodic table, they ensure each element is associated with both a category and a period.

During forward engineering, the logical model, in blue, was converted into a relational model, in yellow. The entities were translated from one table, Element, into relational tables named Element, Category, and Period. Primary keys were created for the identifier attributes, i.e. `atomic_number`, `category_id`, and `period_id`. Moreover, foreign keys were automatically generated in the Element table, which reference the Category and Period tables. Finally, the exported DDL file contains SQL CREATE TABLE statements that represent the relational structure generated from the logical model.